

Source Code and Execution Process

Lightweight MobileViT-Graph Network for Topological Rural Road Extraction

IAMPro-2026 Internship — IEEE CS Bangalore Chapter | Project P29

Repository

GitHub: <https://github.com/Hrithik-Bhattacharya/Satellite-Imagery-based-Road-Network-Extraction>

The repository contains the trained model, all source code, sample satellite tiles, and pre-measured result files. Steps 1 to 5 below reproduce every result in the internship report.

Step 1 — Clone and Install

```
git clone https://github.com/Hrithik-Bhattacharya/
Satellite-Imagery-based-Road-Network-Extraction
cd Satellite-Imagery-based-Road-Network-Extraction
pip install -r backend/requirements-edge.txt
```

The edge requirements install only ONNX Runtime, OpenCV, NumPy and SciPy. PyTorch is not required for inference.

Step 2 — Run Road Extraction on Sample Tiles

Four 1024 x 1024 satellite tiles from rural India are provided in **data/samples/**. Run the ONNX inference script on each tile:

```
python scripts/predict_onnx.py data/samples/100034_sat.jpg \
--model models/mobilevit_v2.onnx --output out_100034.png
python scripts/predict_onnx.py data/samples/102408_sat.jpg \
--model models/mobilevit_v2.onnx --output out_102408.png
python scripts/predict_onnx.py data/samples/115714_sat.jpg \
--model models/mobilevit_v2.onnx --output out_115714.png
python scripts/predict_onnx.py data/samples/117991_sat.jpg \
--model models/mobilevit_v2.onnx --output out_117991.png
```

Each command prints inference time and road pixel fraction. Output PNGs show the binary road mask overlaid on the satellite image.

Tile	Description	Expected road fraction
100034	Forest road, single track	1.76%
102408	Village with dense street grid	11.37%
115714	Farm roads, sparse	1.41%
117991	Rural town, mixed terrain	2.73%

Step 3 — Verify Model Parameter Count

Install full requirements and run the model benchmark:

```
pip install -r backend/requirements.txt
python backend/scripts/test_model.py
```

Expected output: 1,604,657 trainable parameters. This matches Table 5 in the report.

Step 4 — Reproduce Efficiency Table (Table 6 in Report)

```
python scripts/paper_figures/benchmark_efficiency.py
```

Times five models on a 1024 x 1024 tile using 7 repeated runs. Writes results to **figures/real/measurements/efficiency.json**. Pre-measured results are already committed in that file.

Model	Parameters	CPU latency (s/tile)
U-Net	31,037,633	14.75
D-LinkNet34	31,096,129	2.48
DeepLabV3-MBv3	11,020,337	1.00
LR-ASPP-MBv3	3,218,138	0.49
MobileViT v2 (ours, PyTorch)	1,604,657	1.89
MobileViT v2 (ours, ONNX)	1,604,657	0.91

Measured on Intel Core i5-1035G4, CPU only, batch size 1.

Step 5 — Reproduce Pipeline Figures (Figures 4 to 7 in Report)

```
python scripts/paper_figures/local_figures.py
```

Runs the ONNX model on all four sample tiles and generates figures in **figures/real/**. Key measured outputs on tile 117991: connected road components reduced from 9 to 3; 5,076 pixels bridged across canopy gaps by the graph postprocessing step.

Step 6 — Accuracy Results (Table 7 in Report)

Full accuracy was measured by running **notebooks/evaluate_for_paper.ipynb** on Kaggle (Tesla T4 GPU) against all 622 tiles of the DeepGlobe Road Extraction validation split at native 1024 x 1024 resolution. The raw per-tile JSON is committed at **figures/real/measurements/eval_results.json**.

Configuration	IoU	F1	Relaxed F1	cIDice	Topo Prec.	Topo Sens.
Single pass, threshold 0.5	0.515	0.680	0.792	0.767	0.738	0.798
Single pass, full pipeline	0.460	0.630	0.750	0.751	0.679	0.840
4-flip TTA, threshold 0.5	0.536	0.698	0.808	0.778	0.753	0.805

Relaxed F1 counts a predicted pixel correct if it falls within 3 pixels of any ground truth road pixel (Mnih & Hinton 2010 tolerance for thin road detection). Results are on the validation split; split verification showed a 2% precision mismatch so results may be marginally optimistic. TTA improvement over single pass is statistically significant (Wilcoxon p = 0.0007).

Pre-measured Result Files

All numbers in the report are traceable to committed JSON files. These can be inspected without running any script:

File	Contents
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figures/real/measurements/efficiency.json	Latency and parameter counts for all 5 models
figures/real/measurements/onnx_parity.json	PyTorch vs ONNX Runtime max diff = 1.76×10^{-7}
figures/real/measurements/local_figure_stats.json	Pipeline stage stats (component counts, bridged pixels)
figures/real/measurements/eval_results.json	Full accuracy results from Kaggle evaluation (622 tiles)
figures/real/deck/overlay_stats.json	Road pixel fractions on all 4 sample tiles

The evaluation notebook (notebooks/evaluate_for_paper.ipynb) used to produce the accuracy results is committed and reproducible. To re-run: upload to Kaggle, attach the DeepGlobe dataset and models/best_model_v2.pth, and run all cells.